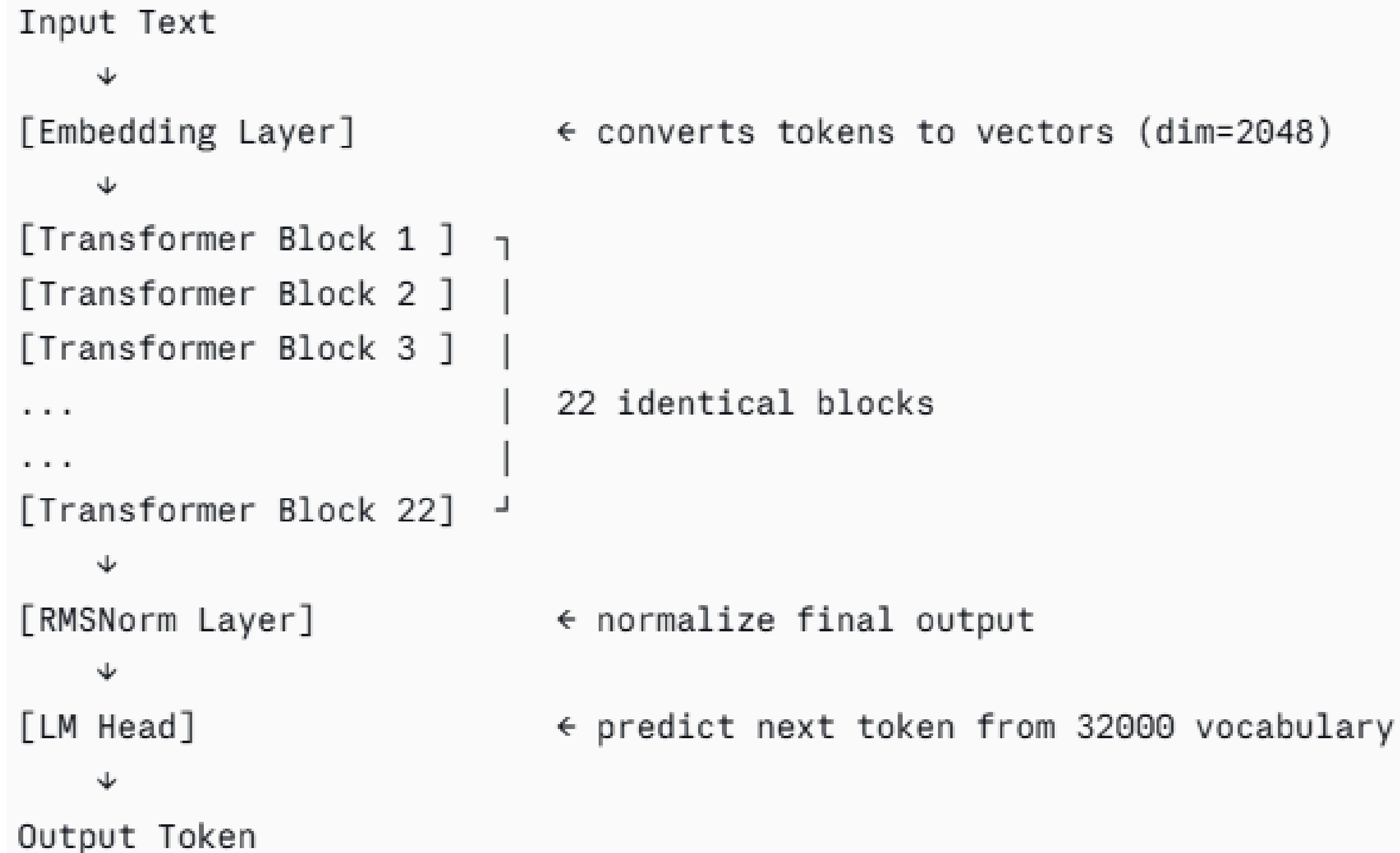




PRESENTATION

By Deva anand M

TINY LLAMA ARCHITECTURE



FORMATED DATA

```
<|system|>
You are an expert legal contract assistant...
<|user|>
What is the Governing Law clause in a legal contract?
<|assistant|>
The governing law shall be the State of Delaware...
```



SYSTEM + USER → ANSWER

Token:	< system >	You	are	legal	...	< assistant >	Two	(2)	years
</s>									
Label:	-100	-100	-100	-100	...	-100	7803	313	29906
2									
	← ignore (input) →					← learn these →			

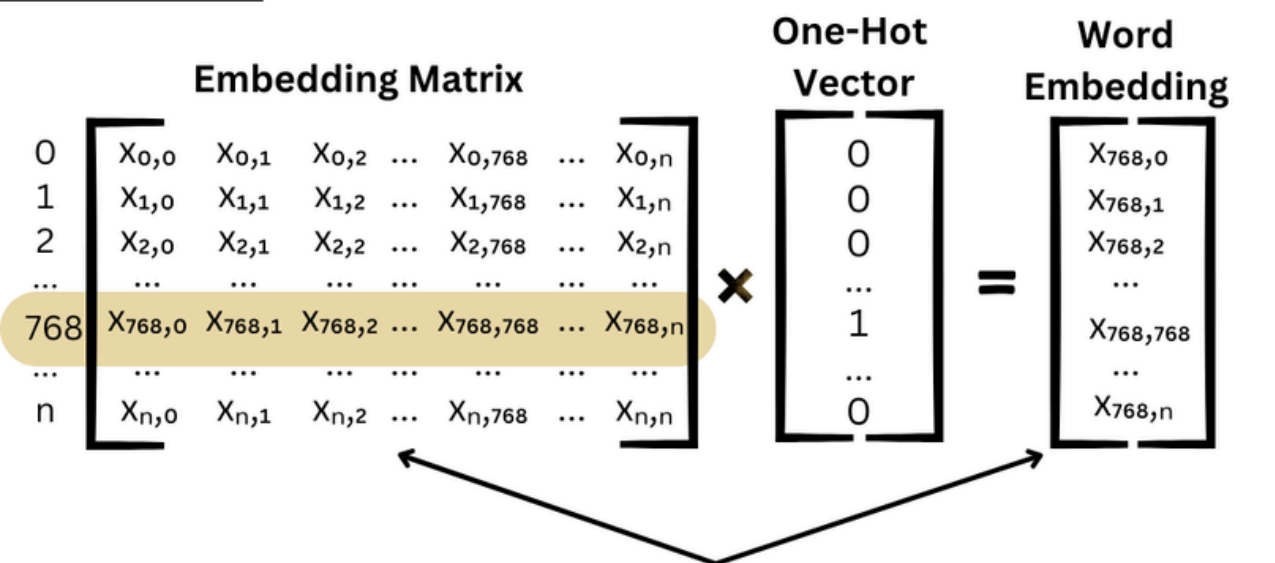
182 prompt tokens	74 answer tokens	126 PAD
Total = 182 + 74 + 126 = 256 ✓ Final shape: [256]		



CONCATENATE & PAD TO 256

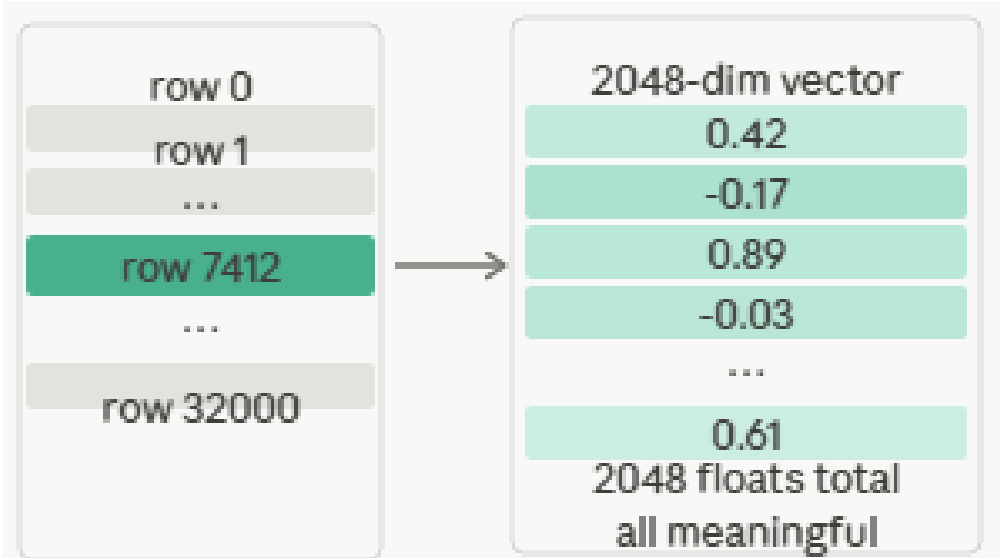
```
[1021, 338, 385, 7412, 29892, ..., 0, 0, 0]
← 256 tokens →
```

Word	king
Token	[768]



The output of this one-hot matrix multiplication is the same as the output of the lookup table!

[1, 256, 2048] TENSOR IS WHAT ENTERS THE FIRST TRANSFORMER LAYER



FOR EVERY BLOCK

SHAPE NEVER CHANGES - ALWAYS 2048 IN, 2048 OUT

[0.234, -0.891, 0.445, 0.123, ...] ← 2048 numbers (meaningless to humans)

↓

LM HEAD
(2048 × 32000 matrix)

↓

Vocabulary scores:
 "the" → 0.001
 "contract" → 0.823 ← highest score
 "agreement"→ 0.412
 "party" → 0.234
 ... 32000 words total

↓

Pick highest score → output word = "contract"

BERTSCORE

Reference: "The cat sat on the mat"

"The" $\rightarrow v_1$

"cat" $\rightarrow v_2$

"sat" $\rightarrow v_3$

"on" $\rightarrow v_4$

"the" $\rightarrow v_5$

"mat" $\rightarrow v_6$

Each v_i is a 768-dimensional vector (BERT output for that word in context)

Prediction: "A feline rested on the rug"

"A" $\rightarrow u_1$

"feline" $\rightarrow u_2$

"rested" $\rightarrow u_3$

"on" $\rightarrow u_4$

"the" $\rightarrow u_5$

"rug" $\rightarrow u_6$

Each u_j is a 768-dimensional vector (BERT output for that word in context)

Cosine similarity formula:

$$\text{cos_sim}(u_j, v_i) = (u_j \cdot v_i) / (||u_j|| \times ||v_i||)$$

Result is between -1 and +1. High value = vectors point in similar direction = similar meaning

Example: $\text{cos_sim}(\text{"feline"}, \text{"cat"}) \approx 0.85$ (very similar)

Example: $\text{cos_sim}(\text{"feline"}, \text{"sat"}) \approx 0.12$ (not related)

PRECISION

$$P = (1 / |\text{prediction}|) \times \sum_j \max_i \text{cos_sim}(u_j, v_i)$$

"FELINE" (u_2) COMPARES AGAINST ALL REFERENCE WORDS. IT MATCHES "CAT" (v_2) WITH SIMILARITY 0.85. THAT'S THE SCORE FOR "FELINE". DO THIS FOR ALL PREDICTED WORDS, THEN AVERAGE.

RECALL

$$R = (1 / |\text{reference}|) \times \sum_i \max_j \text{cos_sim}(u_j, v_i)$$

F1

$$F1 = 2 \times (P \times R) / (P + R)$$



THANK YOU